SIMATS ENGINEERING
ASSESSMENT TOOL 1

COURSE NAME: CSA07 COURSE CODE: Computer Networks

COURSE OUTCOME COVERED

CO1: *Apply the functions of OSI layer in enabling reliable data transmission across networks. (BL2, BL3)*

Assessment Tool Weightage: 40%

Annexure A

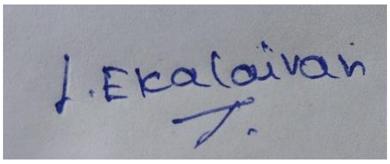
SHORT ANSWER TEST

Dr.V.Parthipan

Code of Conduct:

I EKALAIVAN L (192524450)_(Name / Reg .No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.
I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:



Annexure A

Short Answer Test

1. A company needs to transfer a **150 MB** design file from a client computer to a remote server over a network with a bandwidth of **50 Mbps**. The Transport Layer divides the file into **1500-byte segments**, while the Data Link Layer encapsulates each segment into frames by adding **40 bytes of header and trailer**. Calculate the total number of segments and frames transmitted, the total transmission overhead introduced by the Data Link Layer, and estimate the total transmission time (ignoring propagation delay). Explain how the Transport and Data Link layers ensure reliable communication.

2. A Transport Layer protocol uses a **sliding window size of 6 frames**. Each frame carries **1024 bytes** of user data. If the sender transmits **48 frames**, calculate the number of transmission windows required. If one frame in every window is lost and must be retransmitted, determine the total number of retransmissions. Explain how flow control improves communication efficiency.

3. A **12,000-byte** IP packet is transmitted through a router where the Maximum Transmission Unit (MTU) is **1500 bytes**. Assuming each fragment requires a **20-byte IP header**, calculate the total number of fragments created and the total header overhead introduced. Explain how the Network Layer ensures successful delivery of fragmented packets to the destination.

4. A Transport Layer protocol uses a **sliding window size of 6 frames**. Each frame carries **1024 bytes** of user data. If the sender transmits **48 frames**, calculate the number of transmission windows required. If one frame in every window is lost and must be retransmitted, determine the total number of retransmissions. Explain how flow control improves communication efficiency.

5. A multimedia application transfers a **600 MB** video file. The Session Layer inserts a synchronization checkpoint after every **60 MB** of transmitted data. During transmission, a failure occurs after **390 MB** has been transferred. Determine the number of checkpoints created before failure and calculate the amount of data that must be retransmitted after recovery. Explain the importance of synchronization in the Session Layer.

6. Before transmission, the Presentation Layer compresses a **900 MB** file by **40%**. The compressed data is encrypted, adding **5%** overhead to the compressed size. Calculate the compressed file size, the encrypted file size, and the total percentage reduction compared to the original file. Explain how the Presentation Layer improves transmission efficiency and security.

7. An FTP application transfers **3 GB** of data over a network with an effective throughput of **120 Mbps**. The Application Layer divides the data into requests of **3 MB** each. Calculate the total number of requests generated and estimate the total transmission time in seconds. Discuss how the Application Layer provides reliable services to users.

8. A packet experiences the following processing delays at different OSI layers: Application Layer = **4 ms**, Transport Layer = **3 ms**, Network Layer = **5 ms**, Data Link Layer = **2 ms**, and Physical Layer = **1 ms**. The propagation delay is **18 ms**, and the transmission delay is **12 ms**. Calculate the total end-to-end delay experienced by the packet and explain how each layer contributes to the communication process.

9. A network transfers **80 MB** of useful data using frames of **1600 bytes**, where each frame contains **80 bytes** of protocol overhead added by different OSI layers. Calculate the total number of frames required, the total overhead transmitted, and the transmission efficiency as a percentage. Explain how protocol overhead affects network performance.

10. A hospital transfers a patient's medical record of **240 MB** through an IoT-enabled healthcare network. The Presentation Layer compresses the data by **30%**, the Transport Layer divides the compressed data into **1400-byte segments**, and the Data Link Layer adds **60 bytes** of overhead to each frame. Calculate the compressed file size, the total number of segments transmitted, the total protocol overhead introduced, and the final amount of data transmitted across the Physical Layer. Explain how multiple OSI layers work together to ensure secure and reliable communication.

RUBRICS FOR CALCULATION

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Understanding of Concepts	25%	Demonstrates a thorough, accurate understanding of concepts	Good understanding with minor gaps	Basic understanding with some misconceptions	Poor understanding; major misconceptions
Explanation	25%	Detailed, well-explained answers with relevant examples	Clear explanation with adequate details	Limited explanation; lacks depth	Poor or unclear explanation
Application	20%	Effectively applies concepts with strong analytical ability	Applies concepts with minor errors	Limited application with weak analysis	No application or incorrect analysis
Organization	15%	Well-structured answers with logical flow and coherence	Organized with minor issues in flow	Basic structure, lacks clarity	Poor organization, incoherent answers

Accuracy	10%	Completely accurate and covers all required points	Mostly accurate with minor omissions	Partially correct with missing points	Incorrect answers with major gaps
Clarity	5%	Clear, neat, professional presentation	Understandable with minor issues	Limited clarity, readability issues	Poorly written, difficult to understand

ANSWERS:

1. File Transfer using Transport and Data Link Layer

Given:

File size = 150 MB = 150 × 1,000,000 = 150,000,000 bytes
Segment size = 1500 bytes
Data Link overhead = 40 bytes/frame
Bandwidth = 50 Mbps
Calculation

Number of segments (frames)

= 150,000,000 ÷ 1500

= 100,000 segments

Total frames transmitted

= 100,000 frames

Total Data Link overhead

= 100,000 × 40

= 4,000,000 bytes (4 MB)

Total data transmitted

= 150 MB + 4 MB

= 154 MB

Convert to bits:

154 × 8 = 1232 Mb

Transmission Time

= 1232 ÷ 50

= 24.64 seconds

Explanation

Transport Layer divides the file into segments, provides sequencing, acknowledgements, error recovery, and reliable delivery.
Data Link Layer creates frames, detects errors using CRC, and controls communication between directly connected devices.

2. Sliding Window Protocol

Given

Window size = 6
Total frames = 48
Calculation

Windows required

= 48 ÷ 6

= 8 windows

One frame lost in each window

Retransmissions

= 8 frames

Explanation

Flow control prevents the sender from overwhelming the receiver by limiting the number of unacknowledged frames. This improves efficiency and reduces congestion.

3. IP Fragmentation

Given

Packet size = 12,000 bytes
MTU = 1500 bytes
IP Header = 20 bytes

Payload per fragment

= 1500 − 20

= 1480 bytes

Calculation

Fragments

= Ceiling(12000 ÷ 1480)

= 9 fragments

Header overhead

= 9 × 20

= 180 bytes

Explanation

The Network Layer fragments packets when needed. Each fragment carries its own header, and the destination host reassembles them using identification, offset, and flag fields.

4. Sliding Window Protocol

Calculation
Windows = 48 ÷ 6 = 8
Retransmissions = 8
Explanation

Flow control synchronizes sender and receiver speeds, reducing packet loss and improving network utilization.

5. Session Layer Synchronization

Given

File = 600 MB
Checkpoint every 60 MB
Failure after 390 MB
Calculation

Checkpoints before failure

= 390 ÷ 60

= 6 checkpoints

Last checkpoint

= 360 MB

Retransmission required

= 390 − 360

= 30 MB

Explanation

Synchronization checkpoints allow communication to resume from the last saved point instead of restarting the entire transfer, saving bandwidth and time.

6. Presentation Layer Compression and Encryption

Given

Original file = 900 MB
Compression = 40%
Encryption overhead = 5%
Calculation

Compressed size

= 900 × 60%

= 540 MB

Encrypted size

= 540 × 1.05

= 567 MB

Reduction

= ((900 − 567) ÷ 900) × 100

= 37% reduction

Explanation

The Presentation Layer compresses data to reduce transmission size and encrypts data to ensure confidentiality and secure communication.

7. FTP Data Transfer

Given

Data = 3 GB = 3000 MB
Request size = 3 MB
Throughput = 120 Mbps
Calculation

Requests

= 3000 ÷ 3

= 1000 requests

Convert to bits

3 GB = 24,000 Mb

Transmission time

= 24,000 ÷ 120

= 200 seconds

Explanation

The Application Layer provides user services like FTP, manages file transfer, authentication, and error reporting.

8. End-to-End Delay

Given
Application = 4 ms
Transport = 3 ms
Network = 5 ms
Data Link = 2 ms
Physical = 1 ms
Propagation = 18 ms
Transmission = 12 ms
Calculation

Total Delay

= 4 + 3 + 5 + 2 + 1 + 18 + 12

= 45 ms

Explanation

Each OSI layer performs specific functions such as data formatting, reliability, routing, framing, and signal transmission, contributing to the total communication delay.

9. Network Efficiency

Given

Useful data = 80 MB = 80,000,000 bytes
Frame size = 1600 bytes
Overhead = 80 bytes

Useful data/frame

= 1600 − 80

= 1520 bytes

Calculation

Frames

= 80,000,000 ÷ 1520

≈ 52,632 frames

Total overhead

= 52,632 × 80

≈ 4,210,560 bytes (4.21 MB)

Efficiency

= (1520 ÷ 1600) × 100

= 95%

Explanation

Protocol overhead consumes bandwidth. Higher overhead reduces transmission efficiency and increases network traffic.

10. Healthcare IoT Data Transfer

Given

Original file = 240 MB
Compression = 30%
Segment size = 1400 bytes
Data Link overhead = 60 bytes
Calculation

Compressed size

= 240 × 70%

= 168 MB

Compressed bytes

= 168,000,000 bytes

Segments

= 168,000,000 ÷ 1400

= 120,000 segments

Protocol overhead

= 120,000 × 60

= 7,200,000 bytes (7.2 MB)

Final transmitted data

= 168 MB + 7.2 MB

= 175.2 MB

Explanation

Presentation Layer: Compresses and secures data.
Transport Layer: Segments data and ensures reliable delivery.
Data Link Layer: Adds frame headers/trailers for error detection.
Physical Layer: Transmits the final bit stream over the communication medium.

These layers work together to provide efficient, secure, and reliable communication.